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NITI Aayog

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SECTOR AT A GLANCE

Fisheries and Aquaculture

Agriculture and Allied Sectors in India

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Fisheries and Aquaculture

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The **'Sector-at-a-Glance'** provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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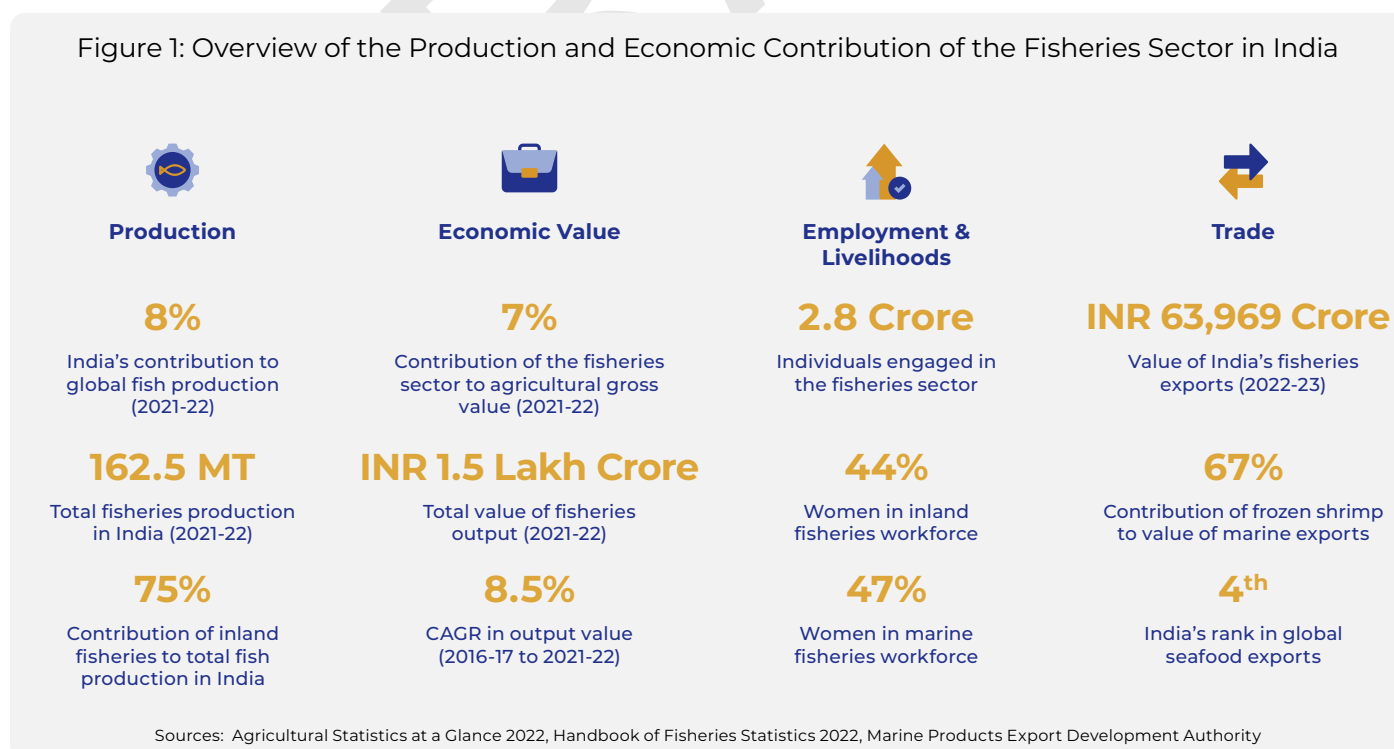
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The fisheries and aquaculture sector makes valuable contributions to the economy and livelihoods of a large portion of the Indian population. The total value of output from India's fisheries sector was INR 1.5 lakh crores in 2021-22, demonstrating the sector's immense economic value to the country. Since 2016-17, India's fisheries production has shown a growth rate of 7% on average, with production in 2021-22 estimated to be nearly 163 lakh metric tonnes. Official estimates suggest that more than 2.8 crore people are engaged in the sector in varying capacities. In terms of global seafood exports, India ranks fourth in the world, exporting goods worth INR 64,000 crores in 2022-23.

Rising demand for fish and other marine products and the integration of technologies, complemented by steady investments by the public and private sectors, are significant tailwinds for this sector. While there is vast untapped potential for fisheries in India, this carries the risk of overfishing and damage to the aquatic ecosystem. The Pradhan Mantri Matsya Sampada Yojana, Sagarmala Programme, and the Fisheries Infrastructure and Development Fund demonstrate the government's commitment to addressing the challenges faced by this sector. These schemes aim to promote sustainable fisheries development and ensure food security for the country.

India's fisheries and aquaculture sector contributes about 8% to global fish production, making India the third largest fish producer in the world. Fish production in India has grown at an average rate of 7% since 2016-17, with the country producing 162.5 lakh metric tonnes (MT) of fish in 2021-22.¹ Inland fisheries are the most important source of fish production in India, accounting for 75% of the total production while the remaining 25% comes from marine sources. Approximately 7% of the total agricultural gross value added (GVA) is contributed by the fisheries and aquaculture sector.² Additionally, the sector provides livelihood to over 2.8 crore people along the value chain of which about 44% are women.¹ India's fisheries exports were valued at nearly INR 63,969 crores in 2022-23 with frozen shrimp contributing to 67% of this total value. India is ranked fourth in global seafood exports as of 2022-23.³

Figure 1: Overview of the Production and Economic Contribution of the Fisheries Sector in India



1 Department of Fisheries. (2022). Handbook on Fisheries Statistics.

2 Department of Agriculture & Farmers Welfare. (2023). Agricultural Statistics at a Glance.

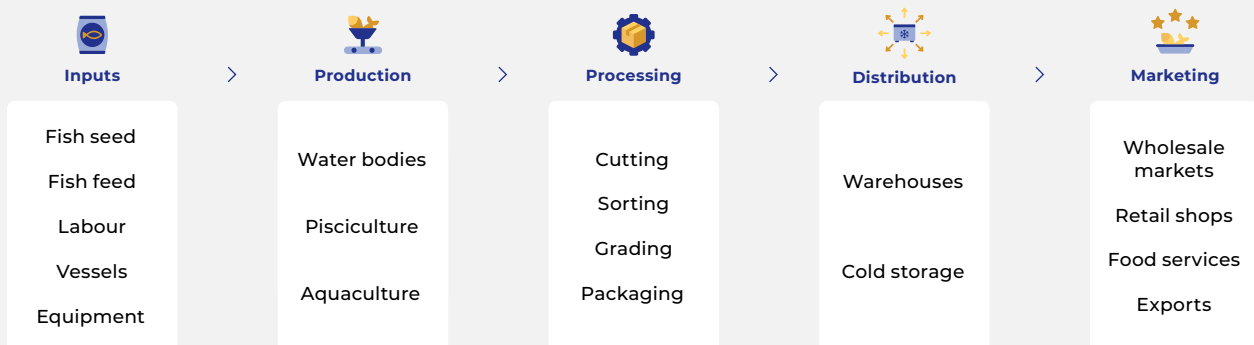
3 Marine Products Export Development Authority. (n.d.). Marine Products Exports. Retrieved on 10 September 2023.

Aquaculture refers to the commercial, recreational, or public cultivation of aquatic species in regulated habitats. Plants and animals are bred, raised, and harvested in ponds, rivers, lakes, the ocean, and artificially constructed “closed” systems on land, among other water settings.

Fisheries Value Chain

The value chain of the fisheries sector in India can be broken down into several stages (Figure 2).⁴ Fish seeds and feed are two main inputs into the sector in addition to labour requirements, fishing vessels, and fishing equipment. Fish seed is an overarching terminology that covers various stages of development in young fish such as hatchlings, spawn, fry, and fingerlings.⁵ Meanwhile, fish feed refers to the food provided to fish and other aquatic fauna that usually includes a combination of nutrients meant to meet or exceed the minimum dietary requirements.

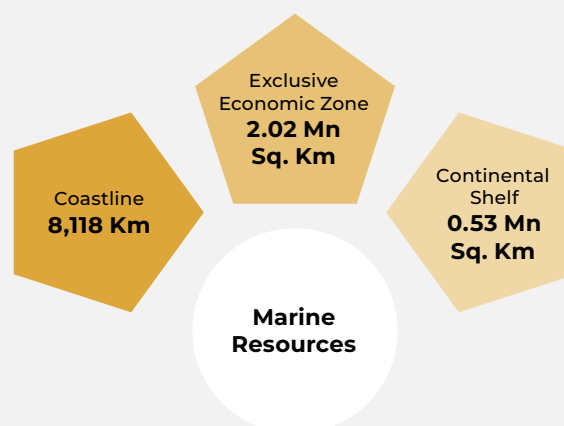
Figure 2: Overview of the Fisheries Sector Value Chain in India



Adapted from Fisheries and Marine Sector (Ministry of Food Processing Industries)

Fisheries production takes place in aquatic ecosystems such as ponds, lakes, rivers, and oceans. With its vast and diverse fishery resources, India contributes more than **10% to the global fish and shellfish biodiversity**.⁶

Figure 3: Marine Resources of India



Source: Department of Fisheries Annual Report 2022-23

The marine resources of the country comprise an Exclusive Economic Zone of 20.2 lakh square kilometres, a continental shelf area of 5.3 million square kilometres and a coastline of 8,118 kilometres (Figure 3). The potential contribution of marine fisheries has been estimated at 53.1 lakh MT.⁶ Additionally, there are 3,461 recognised fishing villages in India.

⁴ Ministry of Food Processing Industries. (n.d.). *Fisheries and Marine Sector*. Retrieved on 15 September 2023.

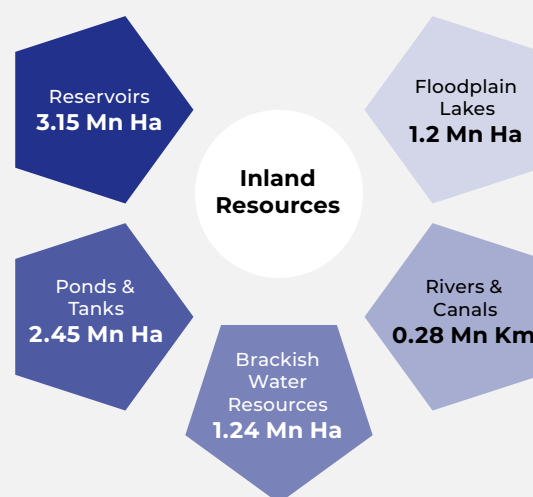
⁵ Food and Agriculture Organization. (n.d.). *Seed Production - An Overview*.

⁶ Department of Fisheries. (2023). *Annual Report 2022-23*.

Inland fisheries consist of 2.8 lakh kilometres of rivers and canals, 12 lakh hectares of floodplain lakes, 24.5 lakh hectares of ponds and tanks, and 31.5 lakh hectares of reservoirs (Figure 4).⁶

Processing of fish products involves a series of activities that result in sorted and graded products that are then distributed using the cold chain logistical network. Adequate infrastructure, including cold storage facilities and transportation, is critical to maintaining the quality and freshness of fish products. The processed fish or fish products are then sold in wholesale markets, retail outlets or restaurants. Alternatively, products are exported to other countries. Exporters purchase fish and seafood products, process them to meet international quality standards, and sell them to foreign markets.

Figure 4: Inland Aquatic Resources in India



Source: Department of Fisheries Annual Report 2022-23

Value Chain Trends

This section of the report discusses the trends in the fisheries sector using a value chain approach and highlights some key data points and facts pertaining to:

1. Fisheries infrastructure (across the value chain)
2. Fish seed production (inputs)
3. Fisheries workforce (inputs)
4. Total, state-wise and origin-wise fish production (production)
5. Production of preserved and processed commodities (processing)
6. Storage and transport infrastructure (distribution)
7. International trade (retail)

Fisheries Infrastructure (Across the Value Chain)

The sector's harvest and handling infrastructure includes fishing vessels, harbours³, and handling centres. As of September 2023, India has over 2.5 lakh fishing vessels of which nearly 55,000 are non-motorised vessels, while approximately 2 lakh are motorised ones. In addition, India has 63 deep-sea fishing vessels, the majority of which are registered in Gujarat (51).⁷ The vessels are complemented by 74 fishing harbours¹, and 225 fish handling centres,⁸ where the latter has a capacity of 2,085 MTs. Processing infrastructure consists of pre-processing plants and processing facilities; India has 605 pre-processing plants with a total capacity of over 11,000 MT and 623 processing plants with a capacity of nearly 36,000 MT.^{9,10} Storage facilities mostly include cold as well as dry storage facilities, amongst other storage premises.¹¹ The total capacity of the storage infrastructure of the country's fisheries sector is estimated at over 5.6 lakh MT.

7 [Marine Products Export Development Authority. \(n.d.\). Fisheries Infrastructure. Retrieved on 20 September 2023.](#)

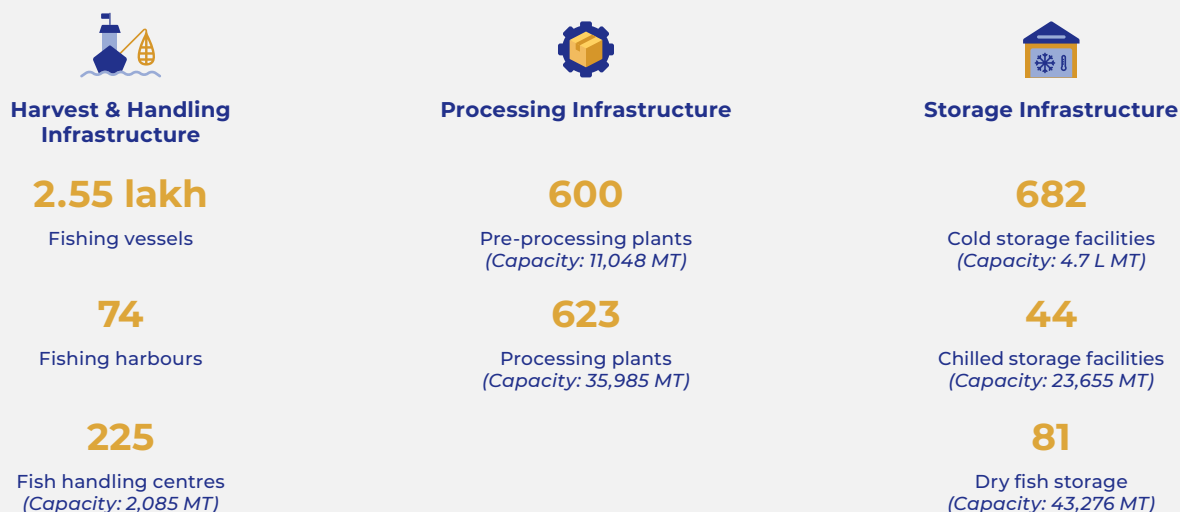
8 [Marine Products Export Development Authority. \(n.d.\). Registered Fish Handling Centres. Retrieved on 20 September 2023.](#)

9 [Marine Products Export Development Authority. \(n.d.\). Region-wise Pre-Processing Plants. Retrieved on 20 September 2023.](#)

10 [Marine Products Export Development Authority. \(n.d.\). Region-wise Processing Plants. Retrieved on 20 September 2023.](#)

11 [Marine Products Export Development Authority. \(n.d.\). Region-wise Storage Premises. Retrieved on 20 September 2023.](#)

Figure 5: Fisheries Infrastructure in India



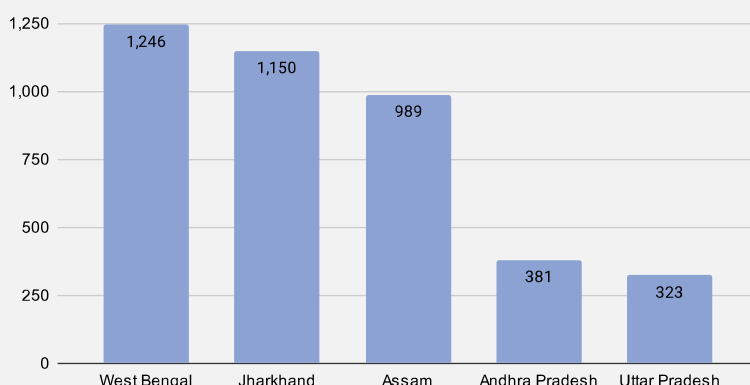
Source: Department of Fisheries Annual Report 2022-23

Fish Seed Production (Inputs)

During the first stage of fish farming, fish seeds are introduced into the system; these can either be sourced from natural habitats or bred on the farm.⁴ The manner of acquiring fish seeds significantly impacts productivity: since inconsistent fish seed supplies can lead to disruptions in other farming operations, a reliable and plentiful supply is necessary for successful fish farming.

Fish seed production in India has grown at a compounded annual growth rate (CAGR) of 8.6%, from 3,57,439 lakh fish fry in 2016-17 to 5,40,690 lakh fish fry in 2020-21.¹ In 2020-21, West Bengal was the largest producer of fish fry (23%), followed by Jharkhand (21%), and Assam (18%) (Figure 6).

Figure 6: Top Five Producers of Fish Seeds (Fish Fry) in India in 2020-21 (Crore)



Source: Handbook on Fisheries Statistics 2022

Fisheries Workforce (Inputs)

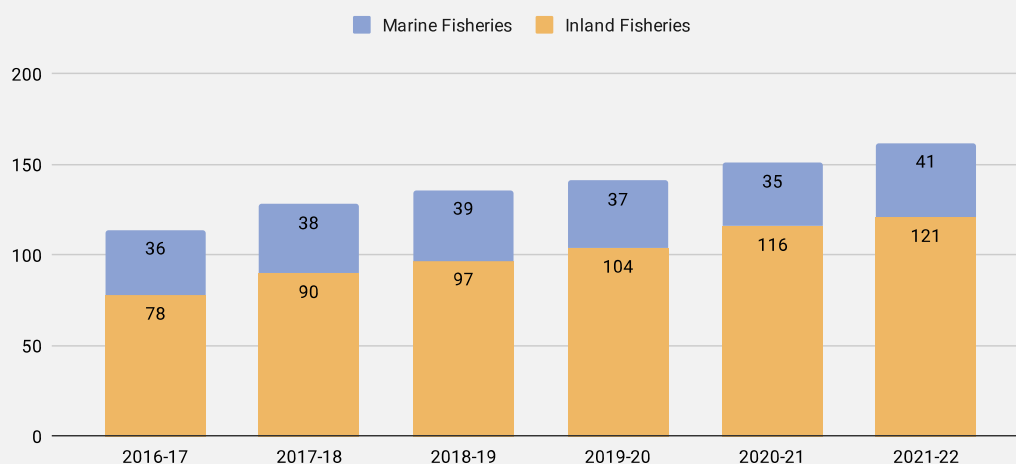
The fisheries and aquaculture sector supports the livelihood of over 2.8 crore people in India. Women are key participants throughout the fisheries value chain; **44% of the inland fisheries workforce and 47% of the marine fisheries workforce comprised of women in 2020-21.**¹ With more than 60 lakh fisherfolk in the state, Bihar has the largest population of the fisheries workforce (22%), followed by Uttar Pradesh (14%), West Bengal (12%), Assam (9%), and Madhya Pradesh (8%). Additionally, as part of the Pradhan Mantri Matsya Sampada Yojana, the government has introduced the new cadre of 'Sagar Mitras' who will act as facilitators to provide extension services to marine fisherpeople in 3,477 coastal villages across the country.¹²

¹² Ministry of Fisheries, Animal Husbandry & Dairying. (2020). Press Release. Retrieved on 11 September 2023.

Fish Production (Production)

Globally, India is the **second largest contributor to the fisheries and aquaculture sector**, accounting for **8% of global fish production**.¹³ Fisheries contributed approximately 7% to India's total agricultural GVA in 2021-2022.² The sector reached a **record high fish production of 162.5 metric tonnes in 2021-22**, with 121.21 lakh MT (75%) accounted for by inland fisheries (75%) and 41.27 lakh MT (25%) by marine fisheries (Figure 7).¹

Figure 7: Fish Production in India (lakh MT)

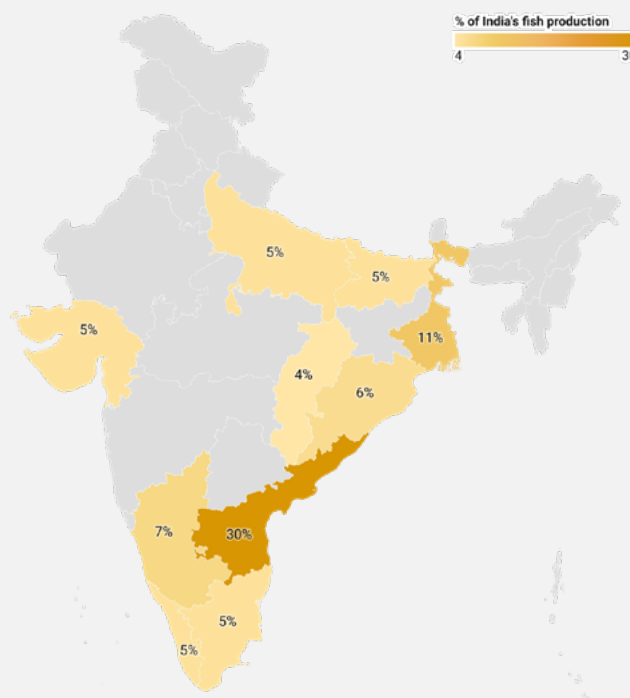


Source: Handbook on Fisheries Statistics 2022

Fish production in India has **grown from 114.4 lakh MT in 2016-17 to 162.5 lakh MT in 2021-22, at a CAGR of about 7%.**¹ The production in the inland fisheries sub-sector has grown at a comparatively faster rate (CAGR of 9%), from 78.1 lakh metric tonnes in 2016-17 to 121.2 lakh MT in 2021-22. In comparison, production from the marine fisheries sub-sector grew from 36.3 lakh MT to 41.3 lakh MT during the same period at a CAGR of 2.6%. **Ten states collectively accounted for approximately 82% of the total fisheries production** in India in 2021-22, with Andhra Pradesh contributing the largest share at 30% (Figure 8).¹

Brackish/saline water aquaculture has witnessed a particularly significant growth, with the production of farmed shrimp recording 7.47 lakh MT in 2020.¹⁴

Figure 8: Top Ten Fish Producing States in 2021-22 (% of Total Fish Production)



Source: Handbook on Fisheries Statistics 2022

¹³ National Fisheries Development Board. (n.d.) *Introduction to Fish and Fisheries*.

¹⁴ Department of Fisheries. (n.d.) *Inland Fisheries in India*. Retrieved on 11 September 2023.

Inland Fisheries¹

121.21 L MT

Total inland fish production
(2021-22)

9.2%

CAGR (2016 - 2022)

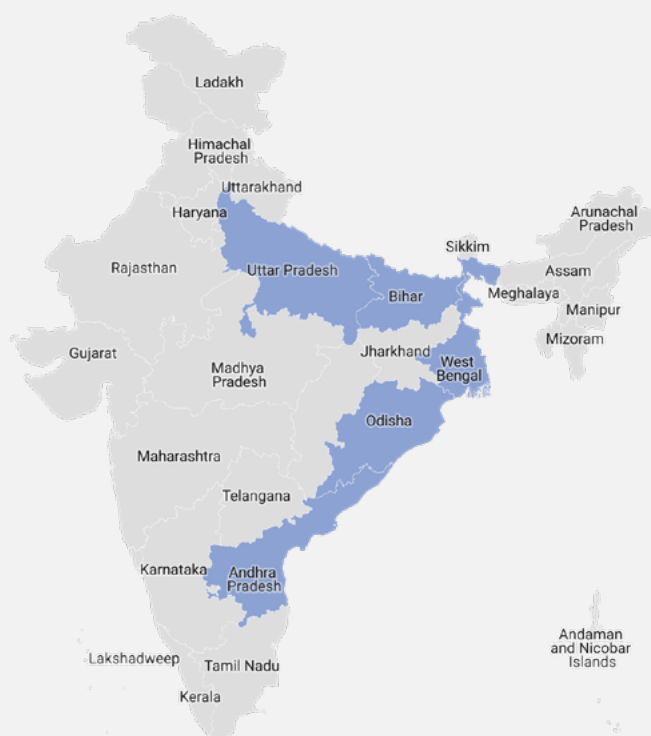
1st

India's rank in inland fish
production

Top 5 Inland Fish Producing
States:

- Andhra Pradesh 42.19 L MT
- West Bengal 16.52 L MT
- Uttar Pradesh 8.09 L MT
- Odisha 7.89 L MT
- Bihar 7.62 L MT

Figure 9: Top Five Fish Producing States - Inland Fisheries
(2021-22)



Source: Handbook on Fisheries Statistics 2022

Marine Fisheries¹

41.27 L MT

Total marine fish production
(2021-22)

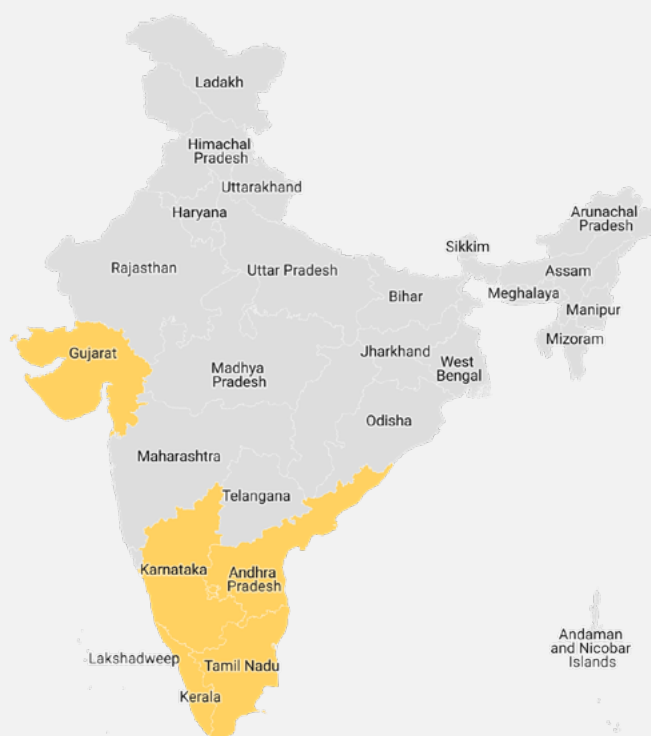
2.6 %

CAGR (2016 - 2022)

Top 5 Marine Fish Producing
States:

- Gujarat 6.88 L MT
- Kerala 6.01 L MT
- Tamil Nadu 5.95 L MT
- Andhra Pradesh 5.94 L MT
- Karnataka 5.89 L MT

Figure 10: Top Five Fish Producing States - Marine Fisheries
(2021-22)

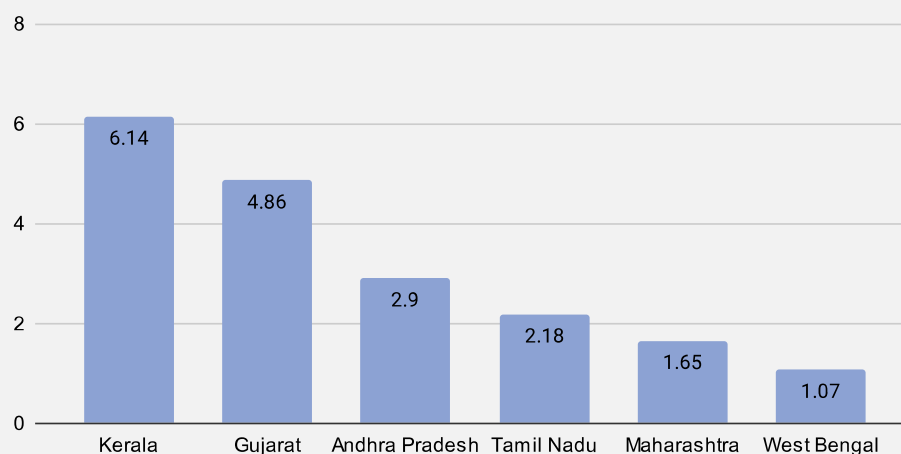


Source: Handbook on Fisheries Statistics 2022

Production of Preserved & Processed Commodities (Processing)

A total of 20.7 lakh tonnes of marine commodities were preserved and processed in India in 2020-21, out of which 90.9% was concentrated in six states (Figure 11).¹ At just over 6.1 lakh tonnes, Kerala is the largest producer of preserved and processed fishery commodities, followed closely by Gujarat (4.9 lakh tonnes). Andhra Pradesh, Tamil Nadu, Maharashtra and West Bengal are the other states producing over one lakh tonnes of preserved and processed fish commodities.

Figure 11: States with > 1 Lakh Tonnes of Preserved and Processed Fish Commodities in 2020-21 (Lakh Tonnes)

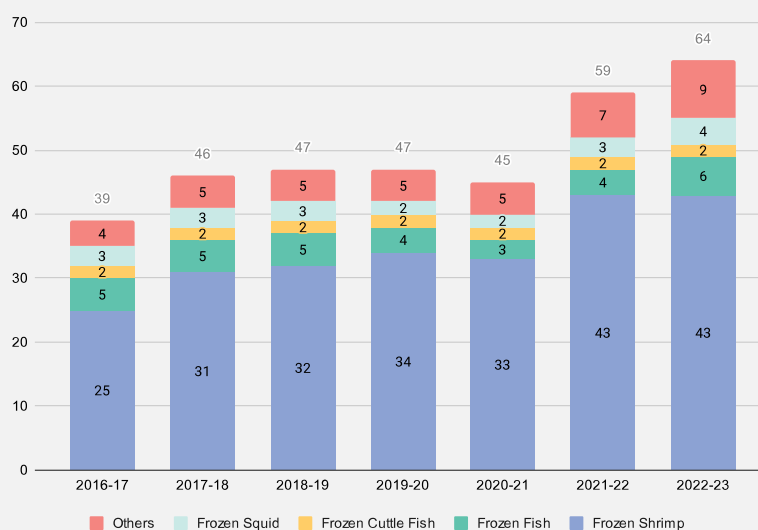


Source: Handbook on Fisheries Statistics 2022

International Trade: Exports (Trade)

India is the fourth largest exporter of fish and fisheries products, valued at INR 63,969 crores in 2022-23. The value of marine product exports increased by over 90% between 2013-14 and 2021-22.¹⁵ **Frozen shrimp** contributed to 67% of the total marine products exports (in value terms) in 2022-23 with a CAGR of 6.9% since 2017-18 (Figure 12). Other marine products exported in 2022-23 include frozen fish (8.6%), frozen squid (5.6%), dried items (4.8%), frozen cuttlefish (3.7%), chilled items (1%), live items (0.7%) and others (8.2%).⁵

Figure 12: Item-Wise Export of Marine Products (INR Thousand Crores)

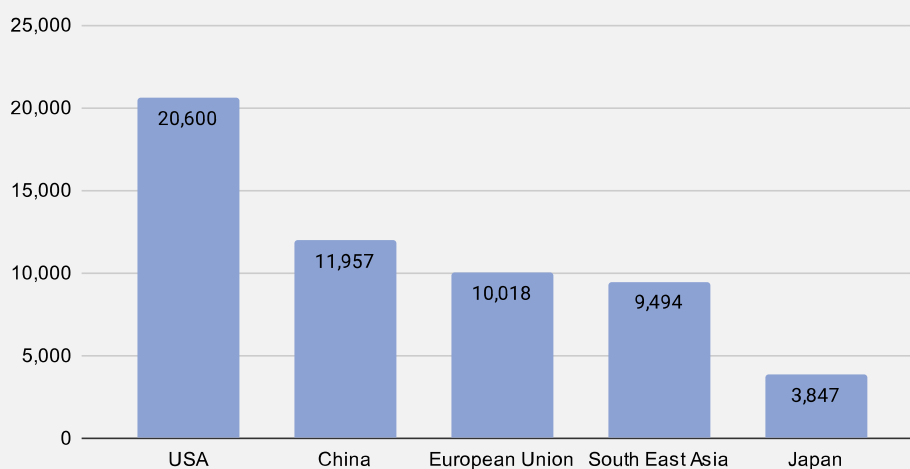


Source: Marine Products Export Development Agency

¹⁵ National Investment Promotion and Facilitation Agency. (n.d.). *Fisheries & Aquaculture*. Retrieved on 11 September 2023.

USA, China, and the European Union are the top three export destinations of Indian marine products (Figure 13), constituting a sum value of over INR 42,500 crores.

Figure 13: Value of Marine Products Exports to Top Five Destinations in 2022-23 (INR Crores)



Source: Handbook on Fisheries Statistics 2022

Opportunities & Challenges

Opportunities



Rising demand for fish products due to affordability and changing food habits:

Consumption and demand for fisheries products are on the rise in both the domestic and the global market, due to their affordability and health benefits. An average of 6.3 kilograms of fish per capita is consumed annually in India; among fish-eating communities, the consumption is even higher at 8-9 kilograms per year.¹



Fisheries offer a high-value livelihood opportunity:

Despite aquaculture being a resource-intensive activity, traditional practices are still retained by some farmers to raise fish with minimum resources. Compared to crops like paddy and vegetables, fish and prawn production generates a net income of INR 0.5-1.5 lakhs per cycle.



Untapped potential of extensive water resources and available marine diversity:

India has about 8,118 kilometres of coastal line; nearly 2 million square kilometres of Exclusive Economic Zone (EEZ); 1.9 lakh kilometres of rivers and canals; and 8.34 million hectares of floodplain lakes, ponds, tanks, reservoirs and brackish water resources.⁶ India is one of the 17 mega biodiversity-rich countries, possessing about 10% of the global biodiversity in terms of fish and shellfish species.



Opportunities for product development in value-added fish products:

There is potential for processing players to innovate Ready-to-Eat and Ready-to-Cook fish products for the domestic as well as the export market, and packaging for increased shelf life and product differentiation.^{4, 6}



Enhancing incomes by focusing on high-value marine species:

Species such as tuna that occur in the oceanic waters are yet to be optimally harvested. There is potential to utilise existing technical resources and knowledge to efficiently harvest such high-value species.



Potential for enhanced foreign investment in the Indian fisheries sector:

100% foreign direct investment (FDI) towards pisciculture and aquaculture is allowed through an automatic route according to India's FDI policy.¹⁶ Therefore, there is immense potential for foreign investment especially for the establishment of farming and processing units.



Underutilised potential of sustainable practices and quality enhancing techniques:

Resource utilisation is optimised and water usage is reduced through practices such as aquaponics, Recirculatory Aquaculture Systems (RAS), high-density pond aquaculture, biofloc systems, and multi-trophic aquaculture. Quality and production can be promoted through genetic improvement programmes such as nucleus breeding centres and brood banks, while intensive and low-cost production can be advanced through pen and cage culture in reservoirs, polyculture, and monoculture. Adequate infrastructure, such as aqua parks, and aquaculture estates/clusters with electricity connection and water inlets/outlets must be developed to facilitate the adoption of these practices.

Challenges^{17, 18, 19}

 Sectoral challenges	• There is slow adoption of technology at production (new hatchery technologies, Recirculatory Aquaculture System, mechanized boats with GPS/satellite tracking systems) and processing stages (value addition technologies). • Limited alignment exists with global standards and certification, such as the Marine Stewardship Council (MSC), affecting export competitiveness; presently, India has only one MSC certified fishery resource – Ashtamudi waters short-necked clam – certified in 2013.
 Input-level challenges	• Aqua-farmers and fishermen face challenges in procuring quality and low-cost aquatic feed. The production of aquafeed in 2014 was only 43.4% of the total feed requirement for aquaculture in the country, forcing India to depend on imports for more than 50% of its requirements.¹⁸ • There exists limited access to extension services, seed and water quality testing facilities, consultations, and other information for fisherpeople and aqua farmers.
 Infrastructural & logistical challenges	• Post harvest losses (physical and quality losses) due to lack of an efficient cold chain infrastructure, with annual post-harvest losses of INR 61,000 crores in fisheries.¹⁹ • Landing and pre-processing centres lack the amenities to maintain required hygiene standards, and small retailers have a limited infrastructure to store goods at appropriate temperatures and environments. • Increased risk of exposure to disease-spreading pathogens is caused due to the lack of knowledge and resources for effective aquatic health management and transport of live aquatic animals and animal products. • Limited access to competitive markets and a poor price discovery mechanism are exacerbated in the unorganised market, compelling small and marginal aquatic farmers to sell their products to intermediaries and dealers at low prices.

¹⁶ Department for Promotion of Industry and Internal Trade. (2020). *Consolidated FDI Policy 2020*.

¹⁷ National Centre for Cold Chain Development. (2017). *Cold Chain and Post Harvest Infrastructure Development Action Plan*.

¹⁸ Vasagam, Ambasankar & Dayal. (2015). *An Overview of Aquafeed Formulation and Processing*.

¹⁹ Keerthana et al. (2022). *Post-harvest losses in the fisheries sector-facts, figures, challenges and strategies*.

Government Intervention

Ministries and Departments

Figure 15: Ministries and Departments Involved in Governance of Fisheries

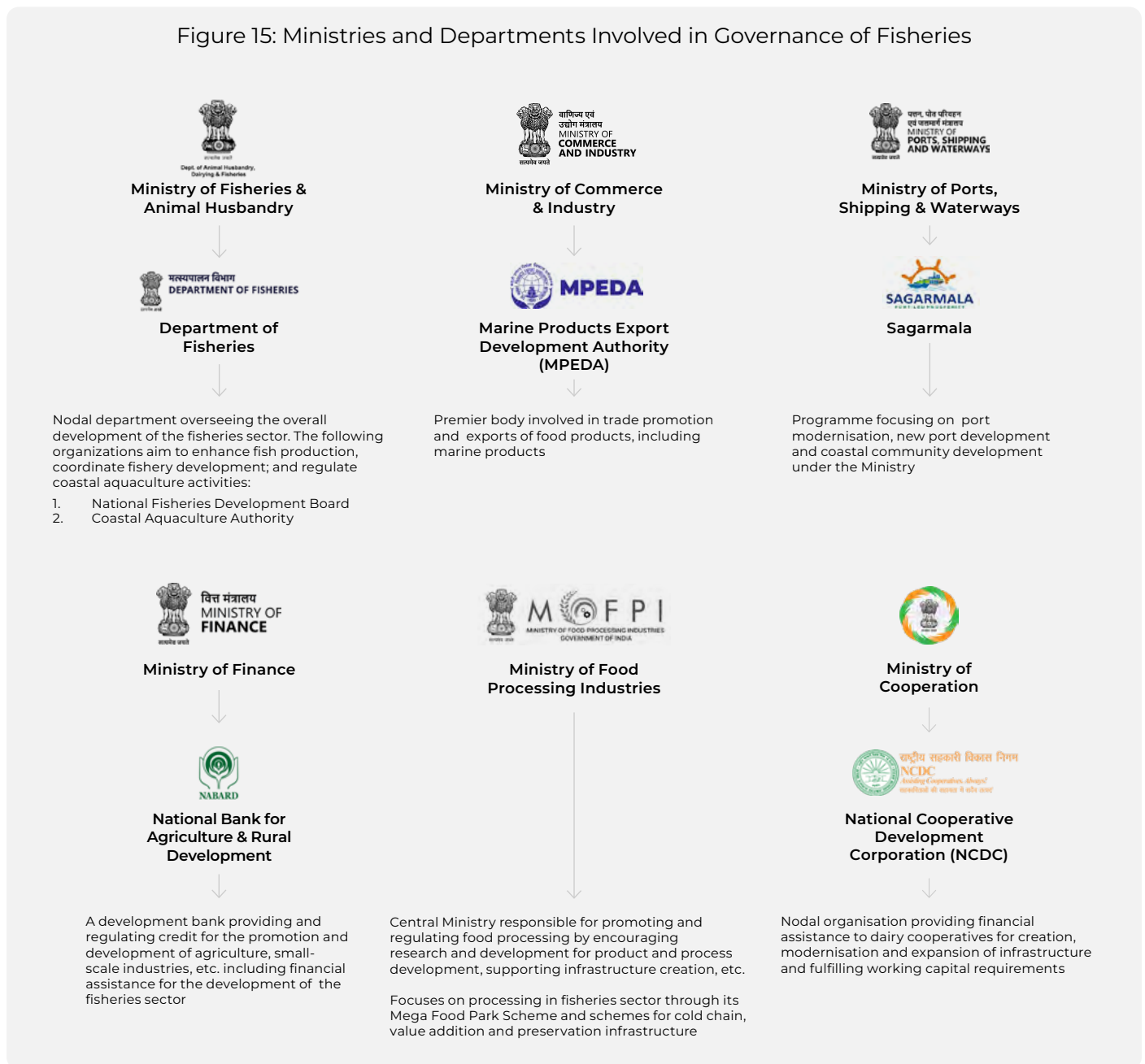


Figure 16: Active Policies, Schemes, and Initiatives in India's Fisheries Sector



SECTORAL INTERVENTIONS

Blue Revolution

Launched during the seventh five year plan (1985-90) with the aim of improving inland and marine production, need-based infrastructure facilities for fisheries, and modernisation in the fisheries sector

Pradhan Mantri Matsya Sampada Yojana (PMMSY)

An umbrella initiative that aims to address critical gaps in the production and quality of fish, post-harvest infrastructure, the value chain, and traceability; intends to establish a robust fisheries management network and strengthen fisherpeople's welfare

National Fisheries Policy 2020

Provides a management and regulatory framework for fish production with the objective of conserving native fish genetic stock, infusing fishing practices with technology, strengthening the value chain with better infrastructure, benchmarking fish products against global standards to enhance trade and exports, and increasing access to institutional credit



INFRASTRUCTURAL AND LOGISTICAL INTERVENTIONS

Sagarmala Programme

Focuses on port-led development, funds fishing harbour projects and reduces logistical costs for domestic and export-import cargo; funds capacity building, infrastructure and social development projects through its Coastal Community Development component

Fisheries & Aquaculture Infrastructure Development Fund

Provides financial support to the state governments and union territories for the development of infrastructure for inland and marine fisheries



INPUT-LEVEL INTERVENTIONS

Kisan Credit Cards for Fisheries

Helps fisheries and farmers meet their working capital requirements through the benefits of its interest subvention and prompt repayment incentives

Pradhan Mantri Matsya Kisan Samridhhi Sah Yojana

Aims to decrease fish feed costs and promote the manufacturing of feed domestically to enhance the earnings of fisherfolk.

Source: Department of Fisheries Publications, Ministry of Ports, Shipping and Waterways Programme Information

20 Department of Fisheries. (n.d.). *Blue Revolution*. Retrieved on 10 September 2023.

21 Ministry of Ports, Shipping & Waterways. (n.d.) *Sagarmala Programme*. Retrieved on 11 September 2023.

22 Department of Fisheries. (2017). *Guidelines for Fisheries and Aquaculture Infrastructure Development Fund (FIDF)*.

23 Department of Fisheries. (2020). *National Fisheries Policy 2020*.

24 Department of Fisheries. (2020). *Pradhan Mantri Matsya Sampada Yojana (PMMSY)*

25 Department of Fisheries. (2021). *KCC for Animal Husbandry, Dairying & Fisheries*.

Way Forward

Looking ahead, the following themes underpinned by the ideals of the 'Blue Revolution' are likely to shape the Indian fisheries and aquaculture sector:



Integrating technology and optimising supply chains

Enhancing the use of technology

- The deployment of artificial intelligence (AI) in fisheries and aquaculture is expected to reduce the intensity of labour and manage activities such as water management, feeding systems, cultivators, and processors.²⁶
- Mobile or blockchain-based seafood traceability systems are to be integrated into the tracking products of the fish marketing system from the source to the point of sale (catch to consumer) in order to maintain public health, ensure high-quality products, and understand environmental impact. Platforms that use blockchain to strengthen the seafood value chain already exist at the global level. Examples include the Provenance project (Indonesia), Fishcoin (global), and Sustainable Shrimp Partnership (Ecuador). All these platforms seek to authenticate and certify products for better transparency, traceability, and quality.²⁷

Leveraging profitability and reducing costs to realise production potential

- The expansion of inland shrimp production can be supported by harnessing the established capacity of hatcheries, feed mills and processing plants. Focus on high-density production of high-value species such as black tiger shrimp can improve profitability.
- According to the ICAR-Central Marine Fisheries Research Institute (CMFRI), the usage of just 1% in-shore waters for open sea cage farming has a production potential of 3.2 million tonnes.²⁸ Therefore, such practices can be promoted to reduce capital costs in aquaculture operations, while also enhancing flexibility in resource use. ICAR-CMFRI has also studied the impact of open sea cage culture in Andhra Pradesh and found a net income increase of 50% to 257% among fish farmers.²⁹
- The development of fish feed mills or farm-made aquafeeds can be supported to enhance sustainability, reduce dependence on imported feed, and decrease feed costs for fish farmers.³⁰



Supporting the fishing community and entrepreneurs

Enhancing income and market access for fishers and aquaculture farmers

- An important focus area of the newly announced Pradhan Mantri Matsya Kisan Samridhi Sah-Yojana (PM-MKSSY) is institutional financing for providing working capital and capital investments. Due to this micro and small businesses would be encouraged to establish supply chains that deliver safe fish products to consumers, enhancing market reach and incomes.
- The development of online marketplaces for e-trading/marketing of fish and other marine products is also supported by the PMKSY.
- Export competitiveness can be improved by aligning the quality of domestic production with global standards and certification, such as the Marine Stewardship Council, etc.

Supporting fisheries collectives through financial and technical assistance

- Primary fisheries collectives and fish farmer producer organisations at the panchayat level must be strengthened in order to increase their bargaining power, viability and diversification of business activities. Their share capital can be built up and production and post-production infrastructure can be improved with financial assistance from the National Cooperative Development Corporation.

Introducing and strengthening a youth entrepreneur-led model for fisheries development and incubation of startups

- Available technical expertise can be enhanced through entrepreneurship incubation. Rapid commercialisation of technologies and local knowledge in fisheries and aquaculture would be enabled, thereby addressing skill gaps throughout the value chain, from input production and transportation systems to processing and marketing.
- As a part of PMMSY, youth entrepreneurship in fisheries can be encouraged through startup challenges and technology incubation centres.³¹

26 Aitwar et al. (2023). *Deployment of Artificial Intelligence in Fisheries and Aquaculture Sector*.

27 Food and Agriculture Organization. (2020). *Blockchain Application in Seafood Value Chains*.

28 Department of Fisheries. (2020). *Sea Cage Farming*.

29 Jeeva et al. (2022). *Success of cage farming of marine finfishes in doubling farmers' income: a techno-social impact analysis*.

30 SAARC Agricultural Centre. (2017). *Farm-made Aquafeeds: Opportunities, Challenges and Policy Interventions*.

31 Department of Fisheries. (2021). *Transforming Fisheries: Pradhan Mantri Matsya Sampada Yojana*. Retrieved on 11 September 2023.



Strengthening infrastructure and skilled human capital

Creating and modernising fisheries infrastructure

- The establishment and modernisation of fish ponds/reservoirs, fish seed hatcheries, marketing infrastructure, mobile transport/refrigerated vans, cold storages, etc. is supported by the Rashtriya Krishi Vikas Yojana and the Fisheries and Aquaculture Infrastructure Development Fund (FIDF).
- Aquatic health management can be improved through the upgradation of laboratory networks.

Supporting food processing in fisheries³²

- Promotion of capacity building is required to enable processing players to prepare value-added products, including traditional and regional items.
- A catalytic role will be played by the government in the support of domestic food processing players (presently, 100% income tax exemption is available to new food processing, preservation and packaging units during the first year of operation, and at the rate of 25%-30% thereafter; and mega food parks are being promoted to enable the setting up of food processing and ancillary units under the Ministry of Food Processing Industries' (MoFPI) Mega Food Park Scheme), as well as foreign players (100% FDI is permitted through an automatic route in the food processing sector).



Promoting the sustainable development of fisheries and aquaculture

Promoting species diversification

- The risk of overfishing of certain species can be reduced and India's export basket can be expanded through species diversification. Ornamental fish breeding can be promoted (such as the native ornamental fishes found in Western Ghats and north-east states of India, and marine ornamental fishes), seaweed cultivation can be improved, and high-yielding species such as tuna can be harvested.

Encouraging sustainable aquaculture practices, habitat preservation and fish stock enhancement

- Organic fish farming can be promoted, the use of chemicals can be avoided and water quality can be maintained; declining fish stocks can be replenished through sea ranching initiatives such as the introduction of hatchery-produced young fish into the sea, and artificial reefs can be created to support diverse fish communities, similar to those found on natural reefs.

Recommended Readings

Food and Agriculture Organization. (2022). *The State of World Fisheries and Aquaculture. Towards Blue Transformation*. <https://www.fao.org/3/cc0461en/cc0461en.pdf>

International Institute for Sustainable Development. (2021). *Supporting Marine Fishing Sustainably: A review of central and state government support for marine fisheries in India*. <https://www.iisd.org/system/files/2021-12/sustainable-marine-fisheries-india-en.pdf>

Kumar, V. M. (2020). Growth and Trade Performance of Indian Fisheries: Trends and Constraints. *National Bank for Agriculture and Rural Development, Mumbai*. <https://www.nabard.org/auth/writereaddata/tender/2501230333growth-and-trade-performance-of%20indian%20fisheries.pdf>

³² Ministry of Food Processing Industries. (n.d.). *Opportunities in Fisheries Sector in India*. Retrieved on 11 September 2023.

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